RideSense AI – Ride Fare Prediction System

RideSense AI, a fast-growing ride-hailing platform, is facing inconsistencies in fare estimation due to changing traffic conditions, weather, peak-hour demand, and dynamic surge pricing. Incorrect fare predictions lead to customer dissatisfaction, driver complaints, and revenue inefficiencies. To solve this problem, the company aims to build a machine learning regression model using 10,000 historical ride records to accurately predict ride fares before trip confirmation. The dataset includes trip distance, duration, traffic intensity, weather conditions, vehicle type, demand levels, driver availability, and surge pricing information. The primary objective is to minimize prediction error using regression metrics such as MAE, RMSE, and R^2 Score while ensuring the model can generalize well under real-world transportation scenarios.

Column Name	Description
trip_distance_km	Total trip distance in kilometers
trip_duration_min	Estimated ride duration in minutes
surge_multiplier	Dynamic surge pricing multiplier
traffic_level	Traffic congestion category
vehicle_type	Type of vehicle selected
weather_condition	Weather condition during ride
peak_hour	Indicates peak-hour trip or not
pickup_zone	Pickup area category
fuel_price_per_liter	Fuel price at trip time
demand_index	Ride demand intensity score
driver_rating	Average driver rating
customer_rating	Average customer rating
music_preference	Preferred in-ride music type
customer_loyalty_score	Customer loyalty membership score
phone_battery_level	Customer mobile battery percentage
ride_fare	Final ride fare amount